

# Stock Price Prediction Using Machine Learning

## Literature Review

### Introduction

Stock trading and investment decisions rely heavily on accurate forecasts of future price movements. With the evolution of data availability and computing power, traditional statistical models are increasingly being replaced by machine learning approaches that better handle complex, nonlinear, and high-dimensional financial data. This literature review outlines the progression from traditional models to advanced machine learning techniques, with a focus on their relevance to stock prediction, and connects these findings with our own analysis.

### Traditional Approaches to Stock Forecasting

Historically, time series models like ARIMA (Autoregressive Integrated Moving Average) and GARCH (Generalized Autoregressive Conditional Heteroskedasticity) have been widely used in financial forecasting (Box & Jenkins, 1976; Bollerslev, 1986). These models are well-suited to capturing linear relationships and volatility clustering but struggle with the non-stationarity and noise typical of stock market data. They also require significant domain expertise and assumptions, limiting their generalizability to complex real-world scenarios.

### Machine Learning in Stock Prediction

Machine learning (ML) techniques have emerged as strong alternatives due to their ability to model nonlinear and dynamic patterns. Supervised learning algorithms, such as Random Forest and Gradient Boosting, have proven effective in various stock prediction tasks. According to Patel et al. (2015), ensemble learning models significantly outperform standalone classifiers in predicting Indian stock indices, citing better adaptability and performance across volatile markets. Gradient Boosting, in particular, has shown high robustness and strong generalization due to its iterative error-correcting mechanism.

In contrast, simpler models like K-Nearest Neighbors (KNN), although computationally lightweight and interpretable, often underperform due to their sensitivity to noise and challenges with feature scaling (Zhang & Zhou, 2007). Despite its limitations, KNN is still frequently used as a benchmark for classification-based prediction systems.

### Importance of Feature Engineering

The success of machine learning models in stock prediction largely depends on effective feature engineering. Technical indicators—especially moving averages such as the 50-day and 200-day averages—are widely used in both trading strategies and ML pipelines (Achelis, 2001). These indicators help reduce market noise and offer a clearer picture of trends, allowing models to learn meaningful patterns. As highlighted in our project, recalculating these averages helped resolve class imbalance issues and significantly improved model performance, particularly for Gradient Boosting.

### Summary of Project Findings

In our project, we analyzed stock price movements for Intel Corporation (INTC) using KNN, Random Forest, and Gradient Boosting Classifiers. We evaluated two different trading strategies and compared model performance based on classification accuracy. Gradient Boosting emerged as the best-performing model with over 90% accuracy under one strategy,

# Stock Price Prediction Using Machine Learning

supporting its effectiveness in prior studies. KNN consistently underperformed, which aligns with existing research regarding its limitations in financial domains.

Additionally, challenges such as skewed data distribution were encountered—an issue well-documented in the literature—and addressed by adjusting moving average thresholds. This not only improved model balance but also reinforced the significance of domain-specific feature refinement in financial ML applications.

## Conclusion

The literature consistently affirms that machine learning models—particularly ensemble techniques like Random Forest and Gradient Boosting—offer improved predictive power for stock market forecasting over traditional statistical methods. Our findings mirror this consensus, demonstrating that Gradient Boosting can effectively capture underlying patterns in price data when combined with informed feature engineering. Moreover, our experience highlights the importance of addressing common issues such as class imbalance, aligning with best practices found in current research. This suggests that careful preprocessing and model selection are critical to building successful trading prediction systems.

## References

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